

Optimal management of cytomegalovirus retinitis in patients with AIDS

Cytomegalovirus (CMV) retinitis is the most common cause of vision loss in patients with acquired immunodeficiency syndrome (AIDS). CMV retinitis afflicted 25% to 42% of AIDS patients in the pre-highly active antiretroviral therapy (HAART) era, with most vision loss due to macula-involving retinitis or retinal detachment. The introduction of HAART significantly decreased the incidence and severity of CMV retinitis. Optimal treatment of CMV retinitis requires a thorough evaluation of the patient's immune status and an accurate classification of the retinal lesions. When retinitis is diagnosed, HAART therapy should be started or improved, and anti-CMV therapy with oral valganciclovir, intravenous ganciclovir, foscarnet, or cidofovir should be administered. Selected patients, especially those with zone 1 retinitis, may receive intravitreal drug injections or surgical implantation of a sustained-release ganciclovir reservoir. Effective anti-CMV therapy coupled with HAART significantly decreases the incidence of vision loss and improves patient survival. Immune recovery uveitis and retinal detachments are important causes of moderate to severe loss of vision. Compared with the early years of the AIDS epidemic, the treatment emphasis in the post- HAART era has changed from short-term control of retinitis to long-term preservation of vision. Developing countries face shortages of health care professionals and inadequate supplies of anti-CMV and anti-HIV medications. Intravitreal ganciclovir injections may be the most cost effective strategy to treat CMV retinitis in these areas.